



MEDIMEI

Evidence-First GenAI Drug Information Q&A Chatbot

Use Case 7 – Q&A Chatbot for Documentation

Team Members:

Anand VB	727823TUAD006
Anbarasu S	727823TUAD008
Gokulkrishnan M	727823TUAD034
Harees Ahamed K	727823TUAD035
Kanishkar P	727823TUAD042
Karthikeyan VS	727823TUAD044
Mithilesh ES	727823TUAD056
Mohanapriyan M	727823TUAD060



Mentor Details

Cognizant Mentor :

Chandramohan K

chandramohan.k2@cognizant.com

College Mentor:

Vimala – k

9865260909

vimalak@skct.edu.in

SKCET



Evidence Grounded



Page Level Citations



Trusted & Transparent

PROBLEM STATEMENT



Healthcare professionals and users often face challenges in getting accurate, trustworthy, and well-cited information from lengthy medical documents such as drug manuals, research papers, and regulatory guidelines.

KEY CHALLENGES

Information Overload

Medical documents are lengthy and complex, making it hard to find relevant answers quickly.

Unreliable Answers

General AI models may generate answers without verifiable evidence or proper citations.

Lack of Transparency

Users cannot verify the source or location of the information in the document.

Time-Consuming

Manually searching and reading documents takes significant time and effort.

Impacts !!

These challenges can lead to delays in decision-making, potential misinformation, and reduced efficiency in clinical and research workflows.

What Users Need?

Accurate Answers

Get precise information from trusted documents.

Evidence & Citations

See exact references with page numbers for every answer.

Trust & Transparency

Verify the source and context of the information.

Fast & Efficient

Get answers quickly and focus on what matters.



Evidence Grounded



Page Level Citations



Trusted & Transparent

USE CASE OVERVIEW



Use Case 7 – Q&A Chatbot For Documentation

Users need prescribing information such as dosage, administration, contraindications, interactions, and precautions for a particular drug. The chatbot must understand the context of these queries and provide accurate information with exact PDF page citations.

WHO WILL USE ?

Healthcare Professionals

Doctors, pharmacists, and clinicians

Medical Students

For learning and reference

Researchers

For evidence-based information

General Users

For trusted drug information

WHAT THE CHATBOT SHOULD DO ?

- Understand user queries related to drug information
- Retrieve relevant information from the provided documents
- Provide accurate and concise answers
- Show exact PDF page citations for every answer
- If information is not found, clearly state that it is not available in the provided document



Evidence Grounded



Page Level Citations



Trusted & Transparent

OUR SOLUTION



MEDIMEI is an evidence-first GenAI drug information chatbot that answers user questions using only the provided medical documents and shows exactly where the answer came from.

How It Works

1. Upload Documents

Upload drug manuals, research papers, or any medical PDFs.

2. Ask a Question

User asks a question in natural language.

3. Retrieve & Understand

System retrieves relevant content and understands the context.

4. Generate Answer

AI generates an accurate answer based on the retrieved evidence.

5. Show Evidence

Answer is shown with exact PDF page citation and source.

KEY HIGHLIGHTS

Evidence-First

Every answer is backed by evidence from your documents.

Page Level Citations

Get exact PDF page numbers for full transparency.

No Hallucinations

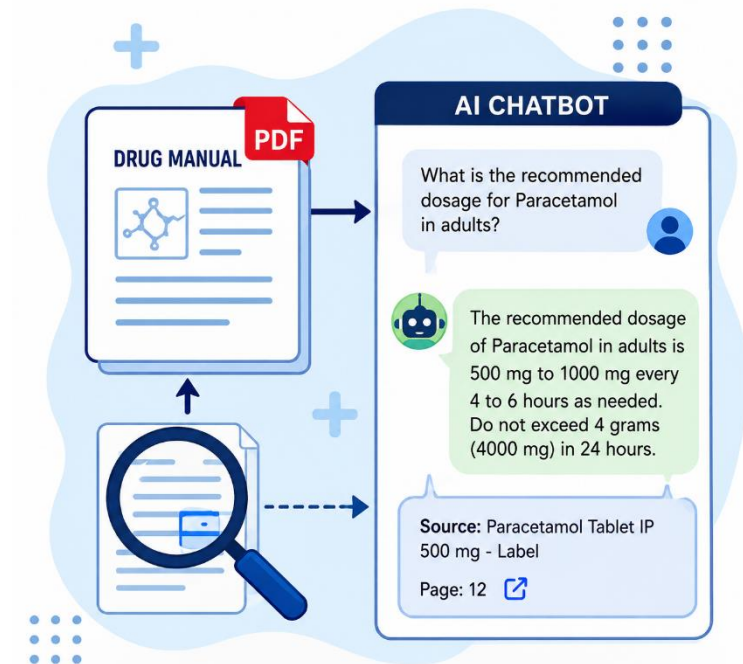
If information is not found, the bot clearly says it.

Secure & Private

Your documents are safe and never shared.

Fast & Reliable

Quick, accurate, and context-responses.



Evidence Grounded



Page Level Citations



Trusted & Transparent

TECHNOLOGIES USED



- **Document Processing**

Extracts text and tables from PDFs, including scanned documents using OCR.

- **Retrieval & Embeddings**

Converts text into vector embeddings and performs similarity search to find the most relevant information.

- **AI Model**

Generates accurate, context-aware answers using retrieved evidence while minimizing hallucinations.

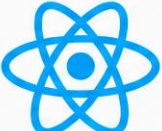
- **Backend & Frontend**

Powers the application, handles queries, displays answers with citations and allows users to open the exact source page.

- **Data & Storage**

Stores user data, metadata, embeddings and documents securely and scalably.

TECH STACK

DOCUMENT PROCESSING   PyMuPDF + PaddleOCR PDF parsing, text & table extraction, OCR for scanned PDFs	EMBEDDINGS & RETRIEVAL   BGE-M3 + Qdrant High-quality embeddings, vector similarity search and storage	AI MODEL  Qwen 3.5 4B Generates accurate, evidence-based answers using retrieved context
BACKEND   Python + FastAPI REST API, query processing, context management, citation generation	FRONTEND   React.js + PDF.js Chat interface, upload documents, show citations and open source pages	DATA & STORAGE   MySQL User data, document metadata, embeddings and files stored securely



Evidence Grounded



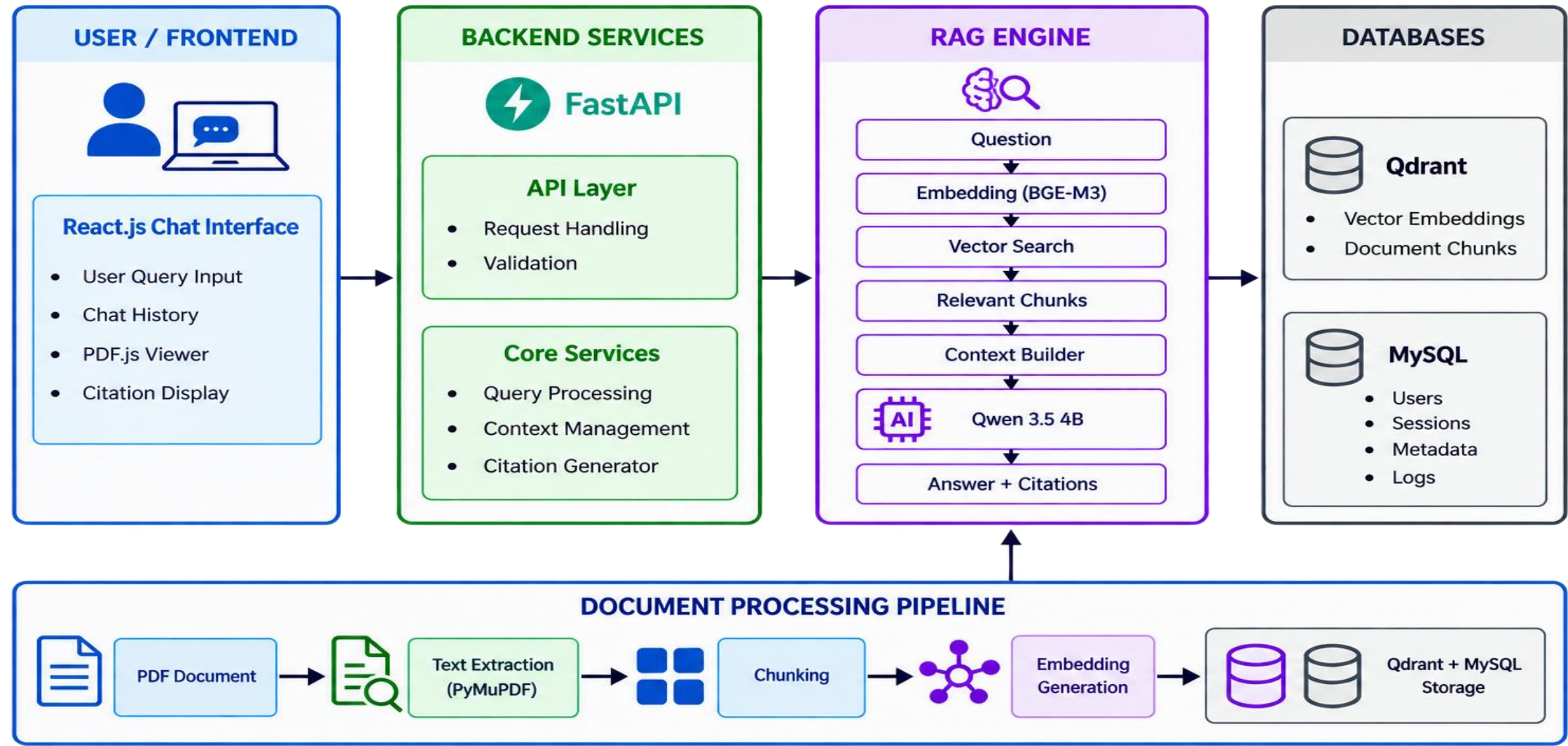
Page Level Citations



Trusted & Transparent



SYSTEM ARCHITECTURE



Evidence Grounded

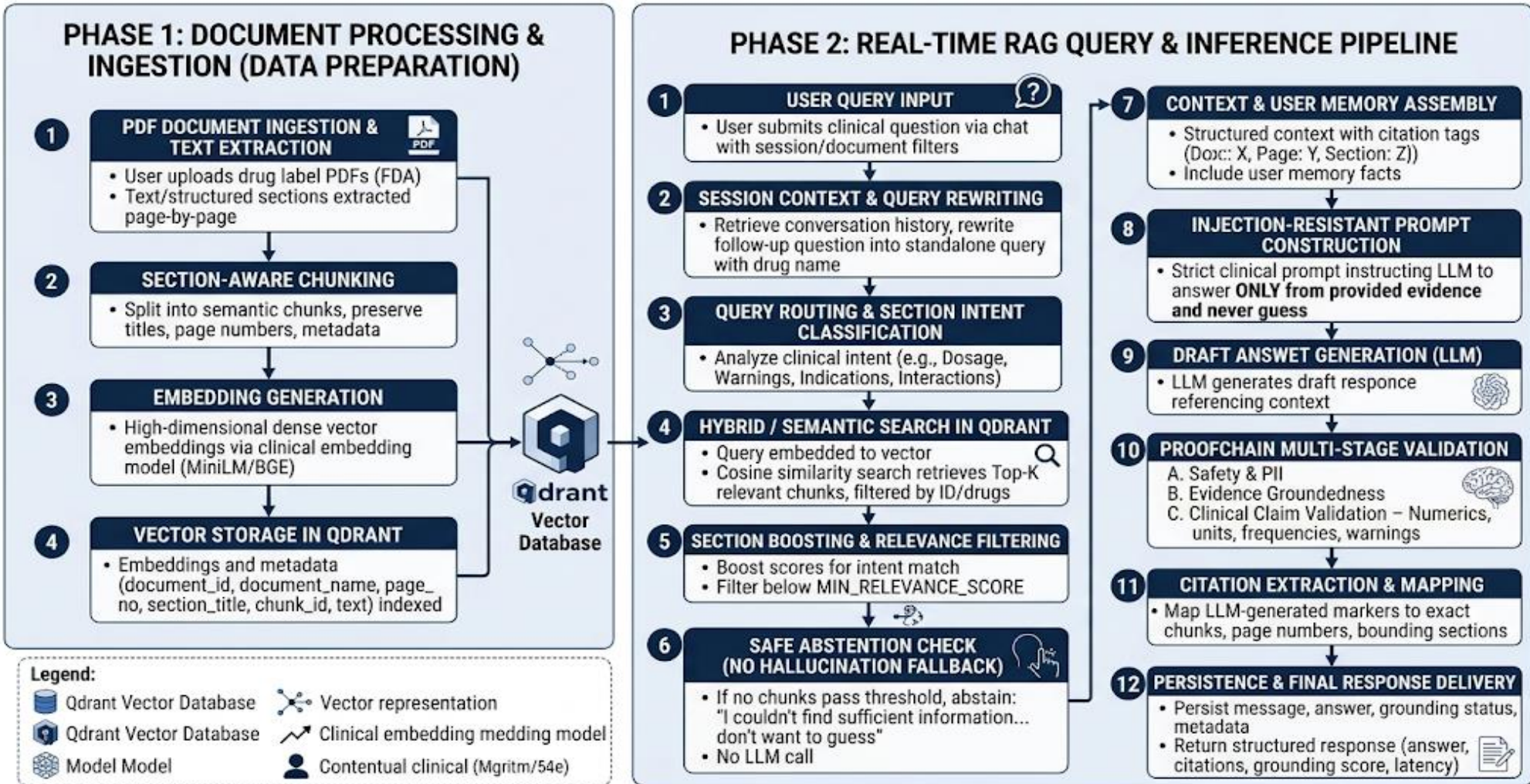


Page Level Citations



Trusted & Transparent

RAG PIPELINE & AI WORKFLOW



Evidence Grounded



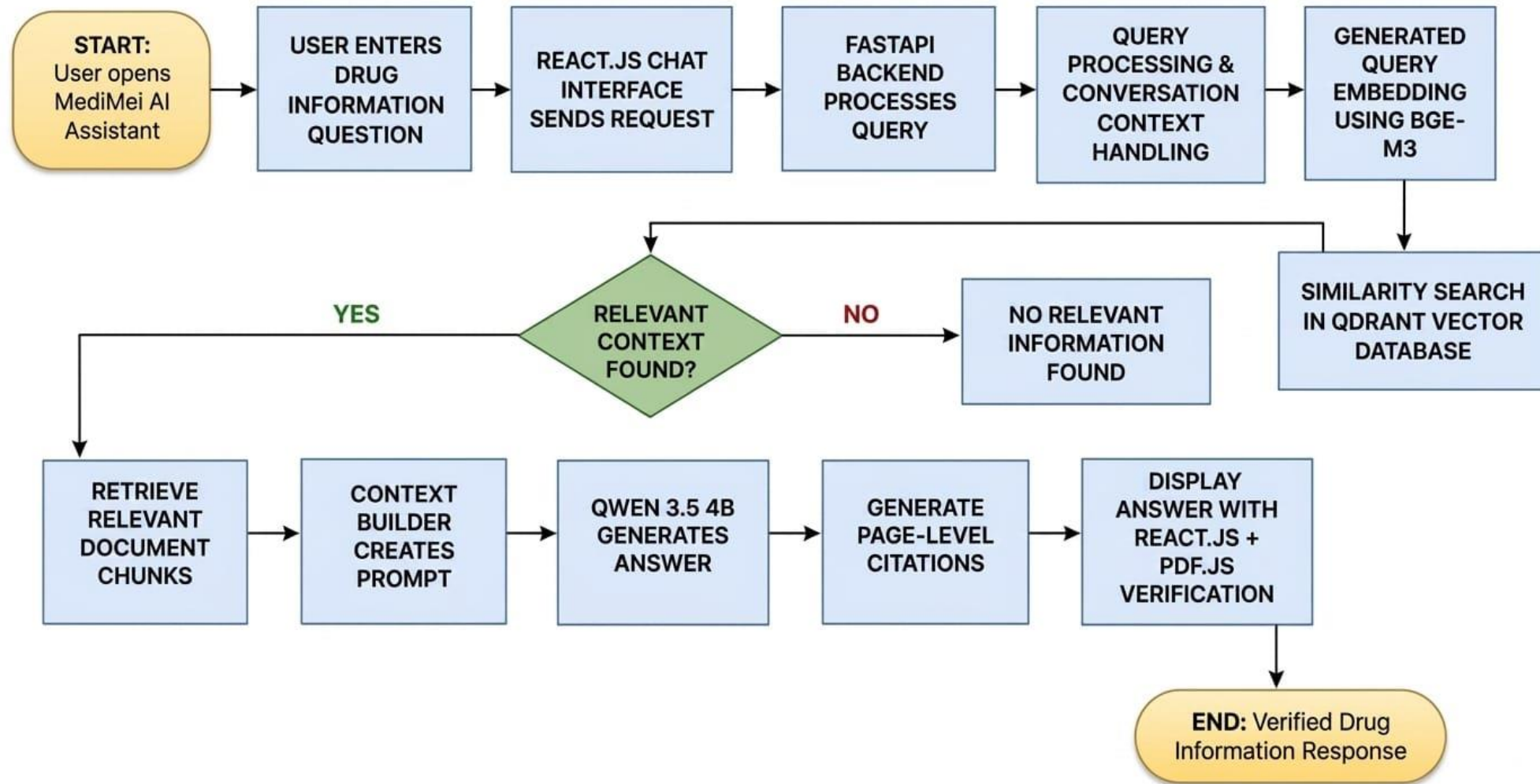
Page Level Citations



Trusted & Transparent



FLOW DIAGRAM



Evidence Grounded



Page Level Citations



Trusted & Transparent

EVALUATION METRICS



Evaluation Area	Metrics	Purpose / What It Measures
Retrieval Quality	Hit Rate@K, MRR, Context Relevance	Measures whether the correct drug-label sections are retrieved.
Groundedness & Relevance	Faithfulness Score, Answer Relevance, Hallucination Rate	Ensures responses are supported by retrieved evidence.
Clinical Safety	Dosage/Unit Accuracy, Warning Recall, Abstention Accuracy	Validates critical medical information and safe responses.
Citation Accuracy	Citation Precision & Recall	Ensures claims are linked to the correct document, section, and page.
System Performance	Retrieval Latency, LLM Generation Time, End-to-End Response Time	Measures overall system speed and efficiency.
Key Benchmarks	>95% Retrieval, 100% Faithfulness, 100% Dosage Accuracy, >98% Abstention & Citation Accuracy, <2.5s Response Time	Defines the target performance standards.



Evidence Grounded



Page Level Citations



Trusted & Transparent

EXPECTED OUTCOMES



- **Near-Zero Hallucination** — ProofChain validates generated answers against retrieved drug-label evidence and prevents unsupported clinical claims.
- **Faster Information Retrieval** — Reduces the time required to find critical information from lengthy drug labels, such as dosage, warnings, contraindications, and adverse reactions.
- **100% Traceable Evidence** — Provides exact document, section, and page-level citations for every evidence-based response.
- **Enhanced Patient Safety** — Identifies critical warnings, contraindications, adverse reactions, and drug interactions from verified label information.
- **Efficient Multi-Drug Comparison** — Enables rapid side-by-side comparison of multiple drug labels for research and clinical evaluation.
- **Privacy-Preserving Intelligence** — Supports local LLM and vector database deployment, minimizing exposure of sensitive clinical information.
- **Safe Abstention** — When sufficient evidence is unavailable, the system refuses to guess and clearly indicates that the information could not be verified.



Evidence Grounded

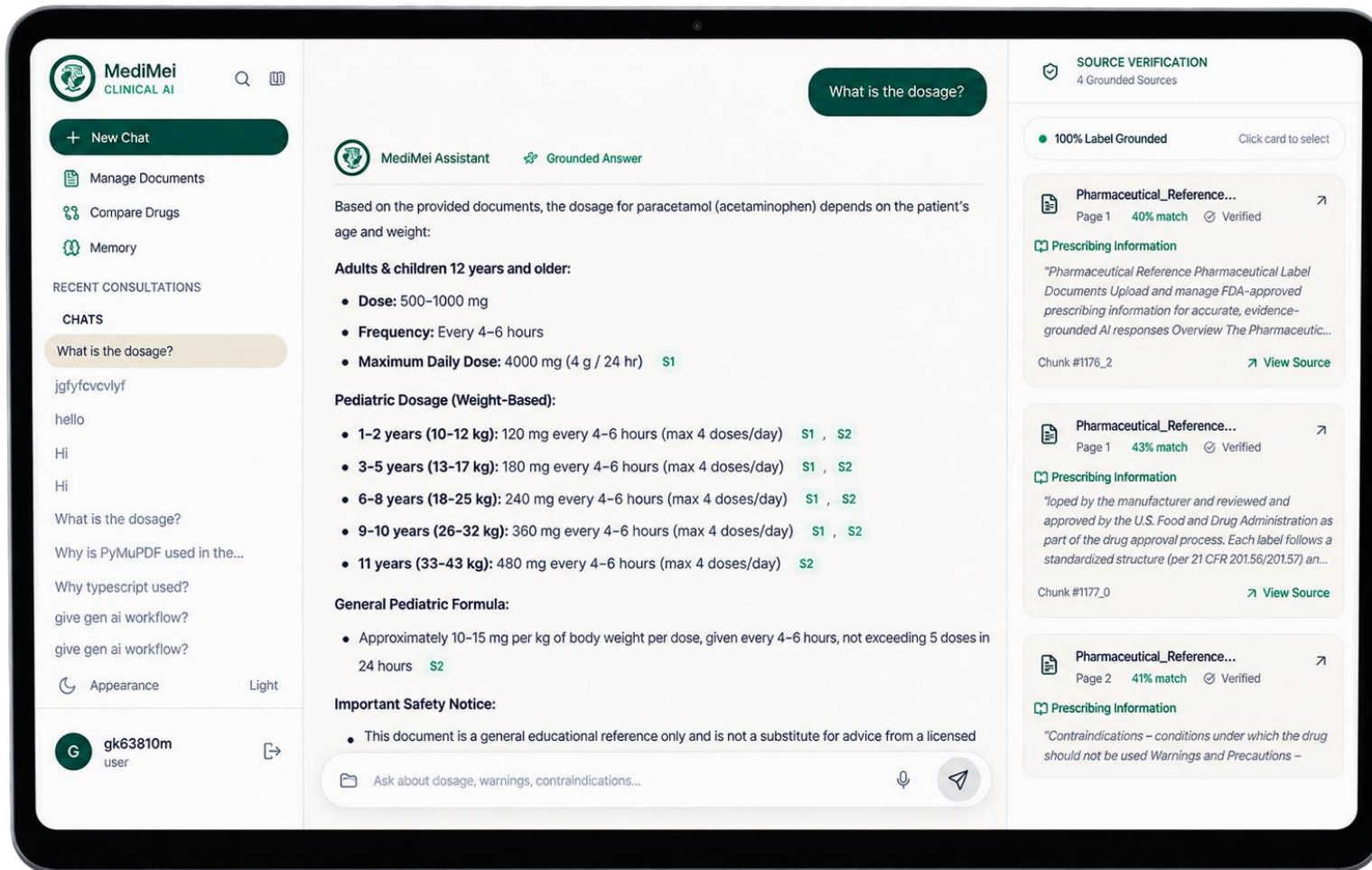


Page Level Citations



Trusted & Transparent

LIVE DEMO & Q/A



MediMei CLINICAL AI

What is the dosage?

MediMei Assistant **Grounded Answer**

Based on the provided documents, the dosage for paracetamol (acetaminophen) depends on the patient's age and weight:

Adults & children 12 years and older:

- **Dose:** 500-1000 mg
- **Frequency:** Every 4-6 hours
- **Maximum Daily Dose:** 4000 mg (4 g / 24 hr) **S1**

Pediatric Dosage (Weight-Based):

- **1-2 years (10-12 kg):** 120 mg every 4-6 hours (max 4 doses/day) **S1**, **S2**
- **3-5 years (13-17 kg):** 180 mg every 4-6 hours (max 4 doses/day) **S1**, **S2**
- **6-8 years (18-25 kg):** 240 mg every 4-6 hours (max 4 doses/day) **S1**, **S2**
- **9-10 years (26-32 kg):** 360 mg every 4-6 hours (max 4 doses/day) **S1**, **S2**
- **11 years (33-43 kg):** 480 mg every 4-6 hours (max 4 doses/day) **S2**

General Pediatric Formula:

- Approximately 10-15 mg per kg of body weight per dose, given every 4-6 hours, not exceeding 5 doses in 24 hours **S2**

Important Safety Notice:

- This document is a general educational reference only and is not a substitute for advice from a licensed

Ask about dosage, warnings, contraindications...

SOURCE VERIFICATION
4 Grounded Sources

- 100% Label Grounded [Click card to select](#)

Pharmaceutical_Reference...
Page 1 40% match **Verified**

Prescribing Information
*Pharmaceutical Reference Pharmaceutical Label Documents Upload and manage FDA-approved prescribing information for accurate, evidence-grounded AI responses Overview The Pharmaceutic...
Chunk #1176_2 [View Source](#)

Pharmaceutical_Reference...
Page 1 43% match **Verified**

Prescribing Information
*Ioped by the manufacturer and reviewed and approved by the U.S. Food and Drug Administration as part of the drug approval process. Each label follows a standardized structure (per 21 CFR 201.56/201.57) an...
Chunk #1177_0 [View Source](#)

Pharmaceutical_Reference...
Page 2 41% match **Verified**

Prescribing Information
*Contraindications - conditions under which the drug should not be used Warnings and Precautions -

01

UPLOAD

Upload the drug PDF document to the system.

PDF



02

ASK

Ask any drug-information question in natural language.



03

VERIFY

Get accurate answer with page-level citation and open the exact source page.



Evidence Grounded

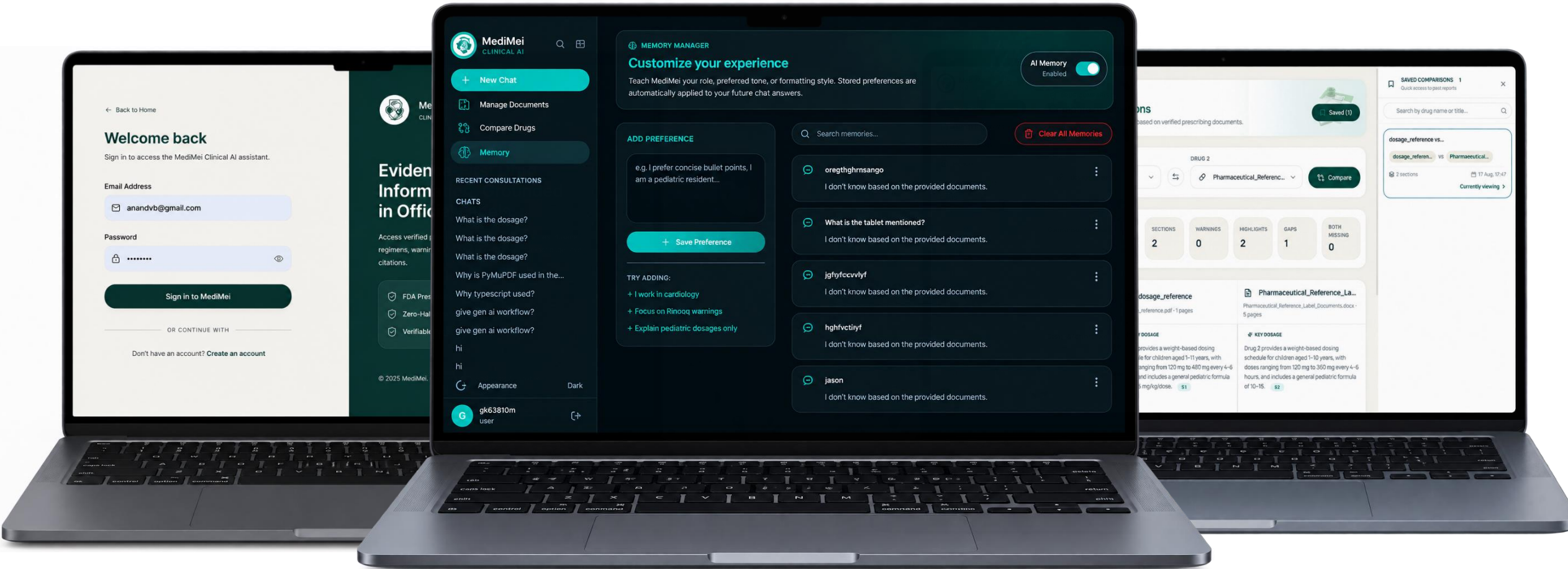


Page Level Citations



Trusted & Transparent

RESULTS



DARK AND LIGHT THEME



Evidence Grounded



Page Level Citations



Trusted & Transparent

DOCUMENTATION



- Hackathon Progress Documentation** — Tracks milestones, development progress, implementation stages, and achievements.
- Individual Contributions** — Highlights each team member's responsibilities, technical work, and completed tasks.
- Model Evaluation** — Measures retrieval accuracy, relevance, groundedness, citation accuracy, and response quality.
- Testing** — Covers functional, integration, retrieval, safety, and end-to-end system testing.
- Performance Analysis** — Evaluates response latency, retrieval efficiency, and overall system performance.
- Results & Validation** — Documents test results, evaluation outcomes, identified issues, and improvements made.
- Focus:** Progress • Contributions • Evaluation • Testing • Performance • Validation



Evidence Grounded



Page Level Citations



Trusted & Transparent

THANK YOU



MEDIMEI

We Don't Guess, We show the proof

